# Spectro-Electrochemical Examination of the Methanol Carbonylation Reaction over Gold

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Dimethyl carbonate (DMC) is an environmentally benign chemical because of its negligible eco-toxicity and low bioaccumulation.[1] It functions as a safe and better carbonylation and methylation reagent to replace the strongly toxic phosgene and dimethyl sulfate for polymer syntheses.[2] Moreover, DMC is also used as polar solvent, electrolyte in energy materials and fuel additive.[1] Compared to the traditional production methods, electrosynthesis uses clean electricity originating from renewables, such as wind or solar, and less waste from spent reagents is generally expected from electrosynthesis.[3] The work aims to shed light on the electrocatalytic DMC production over Au catalysts according to $2CH_3OH + CO \rightarrow (CH_3O)_2CO + 2H^+ + 2e^-$ by means of *operando* attenuated total reflection infrared (ATR-IR) spectroscopy coupled with electrochemistry methods.

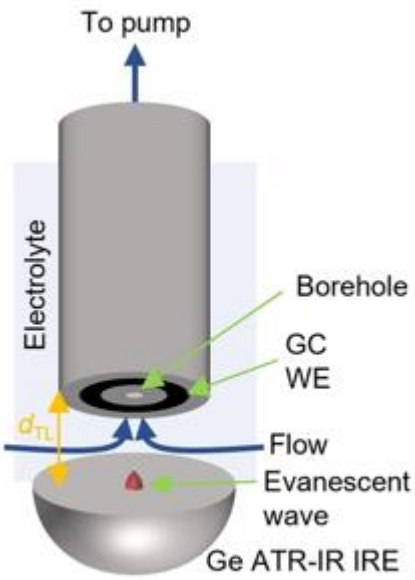


Fig. 1. Illustration of the ATR-IR spectrometer cell.

Fig. 1. depicts the ATR-IR spectrometer cell featuring an Otto configuration with a tilt-correction system, which ensures the parallelly between the IRE and the working electrode (WE).[4] Under the control of a z-approach micrometer screw, the WE is approached perpendicular to the internal reflection unit (IRE) and a thin electrolyte layer with defined thickness ($d_{TL}$) at the interface is formed.[4] By maintaining the potential at -0.3 V (vs. Ag/AgCl (3 M KCl)) for 1 h in the solution of 10 mM $HAuCl_4$ dissolved in 0.1 M KCl, an Au layer is deposited onto the polished glassy carbon (GC) substrate, which is embedded into the borehole ring electrode (BHE). For the methanol carbonylation reaction, a Pt mesh was used as the counter electrode and a homemade Ag/AgCl (3 M KCl) as reference electrode. During the reaction, $d_{TL}$ was maintained at 10 µm and CO gas was constantly bubbled into the electrolyte (0.1 M NaClO4/MeOH) with a flow rate of 10 ml/min.

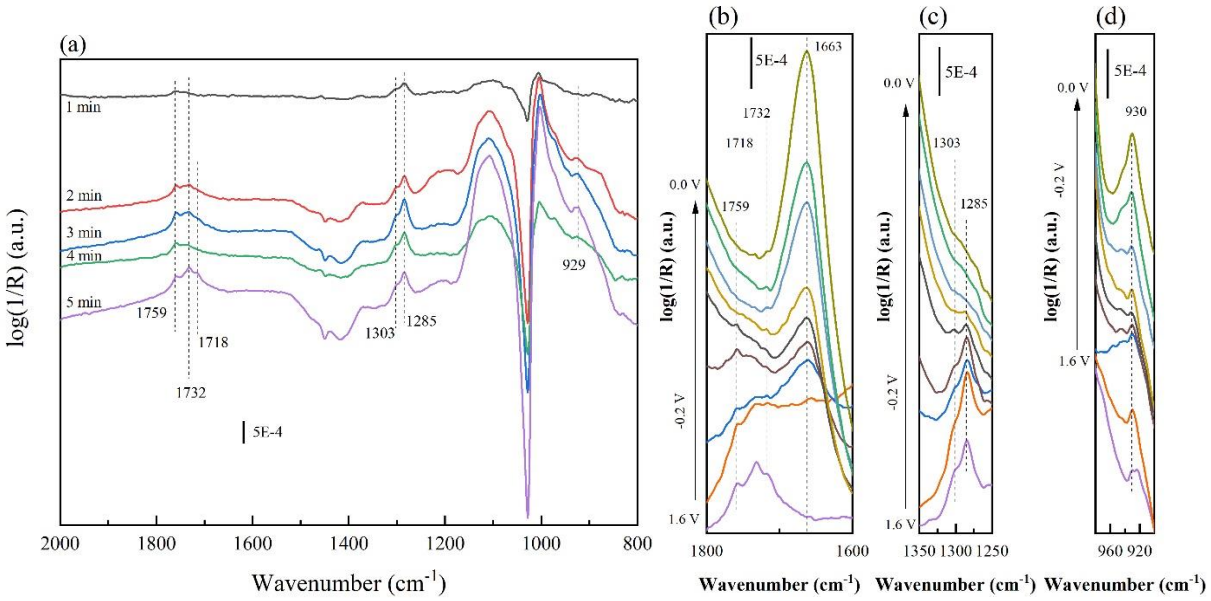


Fig. 2. IR spectra at 1.6 V (a) and all the scanned potentials (b) – (d) during the multi-step amperometry (MA) measurement. MA measurements were started at 1.6 V and scanned to 0.0 V in steps of 0.2 V. Each potential was applied for 5 min.

It is observed in Fig. 2a that at a potential of 1.6 V, DMC (1759, 1303 and 1285 $cm^{-1}$) forms earlier than methyl formate (MF, 1732 and 1718 $cm^{-1}$). However, MF is produced more rapidly than DMC, which makes MF the major product of the reaction. Therefore, further work should focus on suppressing MF formation to improve the selectivity towards DMC. In addition, the MA measurement (Fig. 2b – d) demonstrates that DMC formation is dependent on the potential applied, which implies that DMC formation does not follow a homogeneous pathway. In contrast, dimethoxymethane (DMM, 930 $cm^{-1}$) formation is found to be independent of the applied potential, but depends on time, suggesting a homogeneous process.

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